WawID

Deployment plan

High-availability infrastructure hosted in Senegalese datacenters, backups and disaster recovery

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WAW Telecom · Working document — demonstration data

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This plan describes how WawID moves from the current demonstrator to **sovereign, high-availability** national operations: two active data centers in Senegal, a third backup site (vault), local servers at each border post, a backup strategy, failover and recovery procedures, deployment phases, operations and risks. It is intended for WAW Telecom, the DPAF, Sénégal Numérique, the hosting providers, the DGCPT and the integrators.

It complements `docs/ARCHITECTURE.md` (target national architecture) and the pages *Installation guide — Senegal* (installation of posts, acceptance, go-live), *Deployment and configuration* (demonstrator, variables, NFC bridge) and *Security, personal data and compliance*. The *User guide by profile* describes what users experience in degraded mode and during a failover.

Reading this plan

Label	Meaning
Existing	Works today in the demonstrator (code in the repository).
Target	Architecture selected for national production; it remains to be built.
To be developed	Requires software development absent from the current code.
Recommendation	Advised choice; it commits no party until it has been decided.
To be checked	Fact, certification, capacity or legal point to confirm with the competent party before any decision.

This plan contains no prices, no contractual service levels and no dates: amounts, commitments and schedules will result from consultations and agreements. The durations given are **indicative**.

Summary

Decision	Recommendation
Hosting	In Senegal only, with no foreign cloud for personal data
Site A (primary)	National data center in Diamniadio (State-owned, operated by Sénégal Numérique SA)
Site B (active secondary)	Tier III-certified commercial data center on a separate site and power grid; preferred option: Sonatel (Orange) in Rufisque, alternative: Yas (formerly Tigo) in Diamniadio
Site C (vault)	Separate secure site, outside the Dakar–Diamniadio area if possible, receiving immutable encrypted copies and offline copies
Operating mode	Active-active for stateless tiers (web, API); database in primary/standby with synchronous replication A → B; asynchronous replication to C
Border posts	Local cluster of two servers per post, with a copy of the lists and a local decision engine; checks continue even if both central sites are unreachable
Objectives	Central availability of 99.95% or more; RPO 0 within a site; RPO ≤ 5 min and RTO ≤ 1 h if a site is lost
Backup	3-2-1-1-0 rule, restoration tested every quarter

Decision	Recommendation
Netlify demonstrator	Kept for demonstrations and training, without any real personal data

1. Objectives and requirements

Sovereignty

- **Data and keys in Senegal**: nominative, biometric, travel and financial data, as well as the cryptographic keys that protect it, are hosted and managed on national territory, in accordance with Law No. 2008-12 of 25 January 2008 on the protection of personal data and under the supervision of the CDP. The exact regime (declaration or authorization) is **To be checked** with the CDP.
- **No foreign cloud** for personal data. An encrypted copy abroad would require explicit authorization from the CDP and the State; it is **not recommended**.
- **Key control**: signing and encryption keys are generated and kept in HSMs located in sites A and B, under the control of the State (documented key ceremonies).
- **Reversibility**: the State retains ownership of the data and code; export formats (SQL, objects, audit log) are documented.

Availability and recovery objectives

The values below take up and refine those of `docs/ARCHITECTURE.md`. They are **design objectives**, not contractual commitments.

Indicator	Objective	Remark
Availability of central services	99.95% or more per year	At most about 4 h 20 min of unplanned downtime per year
Data loss within a site (server or rack failure)	RPO 0	Synchronous replication between nodes
Loss of a central site	RPO ≤ 5 min (0 targeted while synchronous replication A → B is healthy); RTO ≤ 1 h	Failover to the other central site
Simultaneous loss of both central sites	RPO according to the last copy received by site C; RTO to be defined (rebuild)	Posts continue checks in offline mode during the rebuild
Checking capacity at the post	100%, including without a link to the central system	Post server (To be developed)
Synchronization of lists to posts	Under one minute	Objective from `docs/ARCHITECTURE.md`
Failover tests	At least twice a year	See section 4

Continuity of checks at posts

Border control must never depend on the availability of the central system. Each post has a **post server** (two-node cluster) that runs the same decision engine as the central system

(`public/shared/engine.js`), keeps an encrypted copy of the watchlists and queues crossings until the link returns ("store and forward"). This mode is **To be developed**: the demonstrator has no offline mode.

Capacity

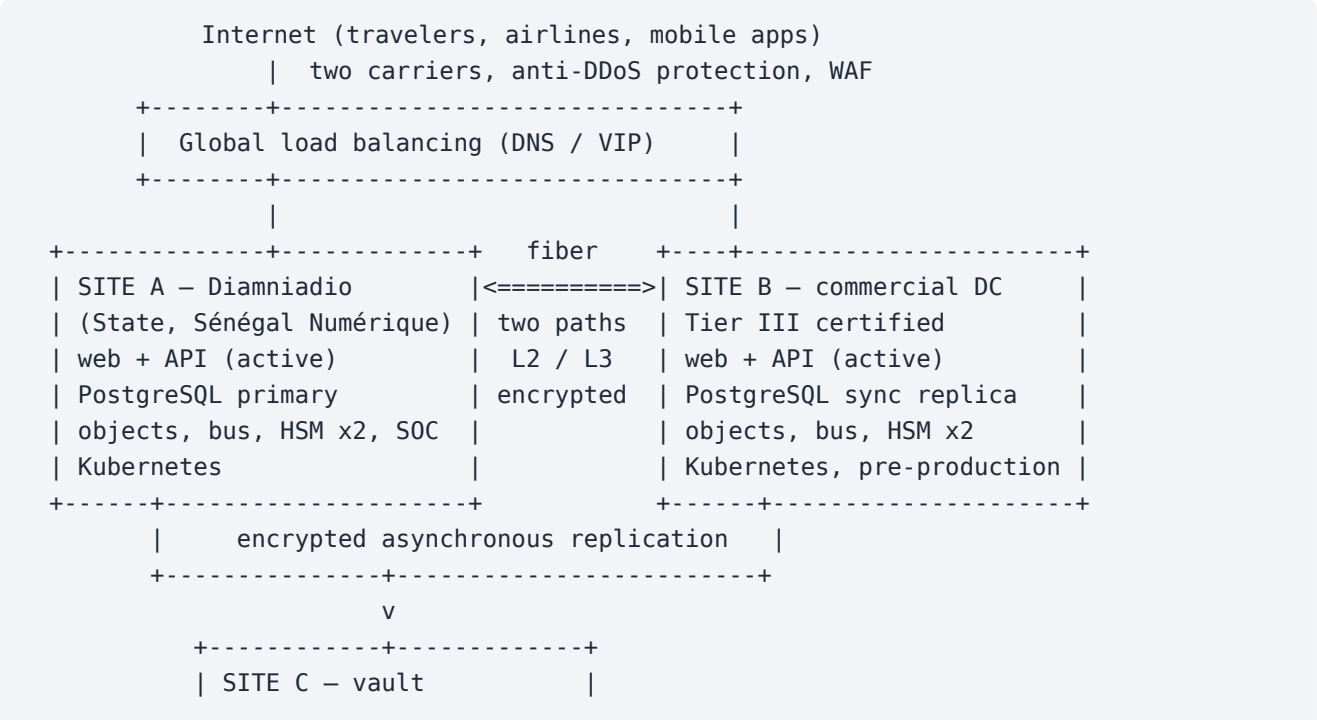
Sizing is based on the assumptions of `docs/ARCHITECTURE.md` , to be confirmed with the DPAF and with AIBD and ANACIM traffic statistics.

Indicator	Estimate	Target sizing (×5)
Annual crossings, all posts	~6 million	30 million
AIBD peak	~3,000 travelers/hour	15,000/h
Checking transactions (national peak)	~2/s	10/s sustained, 50/s burst
List queries	~10/s	50/s
e-Visa applications	~300,000/year	1.5 million/year
Biometric database	~10 million people after 5 years	50 million
Posts, counters, eGates	6 posts, ~60 counters, ~20 eGates	60 posts, 600 counters, 200 eGates
Retention	5 years (configurable)	~150 million crossings

Each central site must be able to carry **alone** 100% of the peak load: in active-active, each site therefore runs at no more than 50% of its capacity in normal conditions.

2. Target topology

Overview



```
           | immutable copies,        |
           | offline copies           |
           +--------------------------+
        MPLS/VPN (carrier 1) + 4G/VSAT (carrier 2)
                        |
  +------+------+------+------+------+------+
  | AIBD | PDK  | ZIG  | ROS  | KID  | KAR  |   post server (2 nodes)
  +------+------+------+------+------+------+   counters, eGates, kiosks
```

Choice of sites

The information below comes from public sources (trade press, data center directories). All of it must be confirmed by the operators during the consultation.

Data center	Operator	Public facts	Status
National data center in Diamniadio	State, operated by Sénégal Numérique SA (formerly ADIE)	Inaugurated in June 2021; about 1,000 m² of technical rooms; 1.4 MW; presented as "Tier III type"	No Uptime Institute certification publicly confirmed: **To be checked**
Yas (formerly Tigo, Sen Connect) Diamniadio	Commercial operator	Uptime Institute Tier III certified (2017), commercial colocation	Nature (design or constructed facility) and validity of the certification **To be checked**
Sonatel (Orange) Rufisque	Commercial operator	Announced as Tier III ("Tier 3+"); second data hall added; expansion (phase 3) reported in 2025 by a single press source; announced proximity to the 2Africa cable landing	Certification, capacity and expansion **To be checked**
Free Sénégal Diamniadio	Commercial operator	Designed to the Tier III level	Not certified according to available information
PAIX Dakar, Onix	Colocation operators	Listed in data center directories	Characteristics **To be checked**

Site A — primary (Recommendation)

The **national data center in Diamniadio** is recommended as the primary site: sovereign State hosting, operated by Sénégal Numérique, consistent with the doctrine on sovereign government files. Points to confirm in the agreement with Sénégal Numérique: available capacity (floor space, power, cooling), actual redundancy level, physical security, access for WAW Telecom teams, available carrier connections, HSM colocation conditions, service levels. No commitment exists to date.

Site B — active secondary (Recommendation)

Site B must be a **Tier III-certified** data center, on a **separate physical site** fed by a **separate electrical substation**, connected through fiber paths different from those of site A, while remaining close enough for synchronous replication (latency in the order of one millisecond).

Criterion	Sonatel (Orange) Rufisque	Yas (formerly Tigo) Diamniadio
Geographic separation from site A	Better: different municipality, distance of roughly fifteen kilometers as the crow flies (actual fiber length to be measured)	Low: same urban hub as site A, shared regional risks (flooding, power substation, road access)
Synchronous replication	Possible (distance under about 20 km)	Possible (very short distance)
Certification	Announced as Tier III, **To be checked**	Uptime Institute Tier III (2017), validity **To be checked**
Power and network separate from site A	Likely, **To be checked**	**To be checked**: risk of a shared substation
Carrier neutrality	Data center of a telecom operator: access for other carriers (second link) must be guaranteed by contract	Same
International connectivity	Announced proximity to the 2Africa cable (To be checked)	To be checked

Recommendation: **Sonatel Rufisque** as first choice for geographic separation, **Yas Diamniadio** as the alternative if the certification and capacity of Rufisque are not confirmed. In both cases, the agreement must guarantee the arrival of at least one carrier other than the data center operator.

Site C — backup vault (Recommendation)

Site C hosts no production. It receives:

- **encrypted, immutable** copies (object storage with write-once locking, or an immutable backup appliance) fed continuously from A and B;
- **offline** copies (encrypted tapes or disconnected disks) renewed every week and kept in a safe;
- backups of the HSM keys (in encrypted form, split among several custodians);
- a lightweight **arbiter** (quorum witness) to prevent A and B from both believing they are primary if the link between them is cut, if site C's link allows it.

Site C should be outside the Dakar–Diamniadio–Rufisque area if possible (another region), in a State facility or a separate secure facility, to be identified with Sénégal Numérique (**To be checked**). No copy is sent abroad.

Operating modes per tier

Tier	Mode	Details (Target)
Web (static files of the platforms)	Active-active A and B	Served by a web tier behind the WAF, in each site
WawID API	Active-active A and B	Stateless containers, at least three instances per site; same code in both sites

Tier	Mode	Details (Target)
Database (PostgreSQL)	Synchronous primary / standby	Cluster managed by Patroni or equivalent; primary in A, local replica in A, at least one **synchronous** replica in B (a transaction is committed only after being written in B); asynchronous replication and continuous archiving of transaction logs (WAL) to C
Object storage (biometrics, documents, exports)	Replicated between A and B	S3-compatible (for example MinIO or Ceph), erasure coding within each site, replication between sites; object locking for archives
Event bus (Kafka or equivalent)	Replicated	Cluster stretched over A, B and an arbiter, or one cluster per site with replication; choice to be validated
Biometrics (ABIS)	Active / standby	According to the capabilities of the selected supplier; galleries replicated in both sites
HSM	Pair per site	Signing keys for QR codes, sessions and audit log anchoring; backup encryption keys; keys replicated through ceremonies
Orchestration	One Kubernetes cluster (or equivalent) per site	Identical deployments driven from a single configuration repository
Identity (OIDC, MFA)	Active-active	Replicated directory and identity provider (To be developed)
Global load balancing	DNS with failover and health checks, or a virtual address announced by both sites (anycast or dual-homing)	Short TTLs; automatic failover when a site's health is lost
Protection	WAF, anti-DDoS, PKI	WAF in each site; volumetric protection provided by the carriers (To be checked); State or dedicated PKI for server, officer and device certificates

Network

Link	Recommendation
Internet for sites A and B	Two different carriers per site (for example two among Sonatel/Orange, Free, Expresso, Yas), separate physical paths; a diversity requirement, without prejudging the choice of carriers
Link A ↔ B	Dark fiber or dedicated layer 2 or layer 3 link, over **two separate physical paths**, encrypted; latency measured before synchronous replication is put into service
Link to C	Dedicated encrypted link, with enough bandwidth for continuous archiving; offline copies are transported physically under a secure procedure
Border posts	MPLS or encrypted VPN to A and B (carrier 1), 4G or VSAT backup (carrier 2); automatic failover
Airlines	Internet, through the WAF, on the `POST /api/carrier/messages` gateway with the `X-Carrier-Key` key (Existing); client certificates (mTLS) considered (Recommendation)

Link	Recommendation
Interpol	I-24/7 connection through the Interpol NCB in Dakar, under the terms defined with Interpol (**To be checked**)
WAPIS, FPR, DAF (civil registry, passports)	Private links or VPN with the departments holding them, after agreements (**To be checked**)
DGCPT / Treasury	Access for Treasury officers through VPN or the administration's private network; encrypted exchange of reconciliation files
Platform administration	Only through a bastion, from the administration network (see section 5)

Exposure and domains

Today, the four platforms are served under a single domain, with separate paths (Existing): DPAF space `/dpaf/`, traveler platform `/`, mobile apps `/app/`, airline platform `/compagnies/`, kiosk `/gate/`. In production, each platform gets its own domain (Recommendation):

Platform	Indicative domain	Exposure
Travelers (portal, help, e-Visa, eGate pre-enrollment)	`www.<domain>` or `voyageurs.<domain>`	Internet, behind the WAF
Mobile apps and their API	`app.<domain>` (page) and `api.<domain>` (public routes)	Internet, behind the WAF
Airlines (platform and gateway)	`compagnies.<domain>`	Internet, behind the WAF, rate limiting
DPAF space	`dpaf.<domain>`	Private network or VPN only, never on the Internet
eGate kiosks	Served by the post server on the post network	Post network only

The code calls the API through relative paths (`/api/...`): each domain therefore routes `/api/*` to the API tier. Links between platforms and shared resources (`/shared/`) must be checked during the migration. Since the session cookie is bound to the domain, separating the domains also separates sessions between platforms.

Border posts (edge)

Item	Target (To be developed)
Servers	Two-node cluster per post (AIBD, PDK, ZIG, ROS, KID, KAR); AIBD capacity to be sized for the peak
Local data	Encrypted copy of the watchlists, approved e-Visas and CSCA certificate lists, synchronized in under one minute
Decision	`public/shared/engine.js` engine run locally; local 1:1 biometric comparison
Crossings	Recorded locally then forwarded to the central system (store and forward), with a unique identifier to avoid duplicates
Attached devices	Counters, eGates, kiosks, CCTV, plate readers; they talk to the post server, which talks to the central system

Item	Target (To be developed)
Power	UPS and generator; autonomy sized according to observed outages
Network	Two links (main and backup), local firewall, segmentation

What does not work without a link: real-time queries to external sources (Interpol, WAPIS), 1:N biometric search, reception of new API/PNR messages, knowledge of e-Visas approved or revoked after the cut. These limits are offset by the local copy and by reconciliation when the link returns.

3. Backup strategy

3-2-1-1-0 rule

Item	Implementation (Target)
3 copies	Production (A), replica (B), backups (A, B and C)
2 media	Disk (backup storage) and removable media (tape or offline disk)
1 off-site copy	Site C
1 immutable or offline copy	Write-once storage at site C and a weekly offline copy in a safe
0 errors	Automatic verification of each backup (checksums, test restoration); any error opens an incident

Schedule

Data	Frequency	Destination
PostgreSQL transaction logs (WAL)	Continuous archiving, point-in-time recovery (PITR)	A, B and C
Full database	Daily	A and B, copy to C
Incremental database	Hourly	A and B, copy to C
Object storage (biometrics, documents)	Continuous replication A ↔ B; versioning; daily copy	C
Audit log	Daily export of the SHA-256 chain and of the chain head signed by the HSM	C, in addition to the database
Configuration (Kubernetes, network, firewall)	At each change, from the configuration repository	A, B and C
Offline copy	Weekly	Safe at site C
HSM keys	At each key ceremony	Encrypted backup split among custodians, at site C and in a second safe

Retention

The data retention period shown in WawID is **5 years** (`retentionDays` setting, 1,825 days, Existing). The applicable legal period is **To be checked** with the CDP and the DPAF. Backups follow the same rule: data purged from production must disappear from backups at the latest when their own retention period expires. The demonstrator applies **no automatic purge**: the purge and archiving job is **To be developed**.

Encryption and keys

- All backups are encrypted before they leave the server that produces them.
- Backup encryption keys are protected by the HSMs (master key never exported in clear).
- Restoration requires two authorized people (four-eyes principle).

Restoration tests

Test	Frequency	Criterion
Full database restoration in pre-production	Quarterly	Consistent database, duration measured, RPO observed
Point-in-time restoration	Quarterly	Return to the requested point in time
Restoration from site C (immutable copy)	Twice a year, during drills	Rebuild possible without A or B
Verification of the restored audit log	At each restoration	**Chain intact** and chain head consistent with the last HSM signature

4. High availability and disaster recovery

Failure scenarios

Scenario	Detection	Decision	Target RTO	Target RPO	Main actions
Loss of a node (server, container)	Automatic monitoring	Automatic	A few minutes	0	Restart or takeover by the other instances; node replacement
Loss of a rack or room within a site	Monitoring	Automatic, then operations manager	Under 15 min	0	Database failover to the local or remote replica; capacity absorbed by the other site
Loss of site A	Monitoring and arbiter	Automatic for the database if the arbiter confirms; otherwise decision by the crisis unit	≤ 1 h	≤ 5 min (0 if synchronous replication is healthy)	Promotion of the B replica; DNS or virtual address to B; audit log check; posts informed

Scenario	Detection	Decision	Target RTO	Target RPO	Main actions
Loss of site B	Monitoring	Automatic	No outage (A remains primary)	0	Synchronous replication switches to degraded mode (asynchronous to C); alert; B restored
Both central sites unavailable	Post monitoring	Crisis unit	Rebuild: to be defined	Last copy received by C	Posts in offline mode; rebuild from C; resynchronization of queued crossings
Link loss at a post	Post server, monitoring	Automatic	Failover to the backup link within a few minutes	0 (queued crossings)	4G/VSAT backup; otherwise offline mode; resynchronization when the link returns
HSM failure	Monitoring	Automatic	No outage (twin HSM)	0	Replacement; key resynchronization through a ceremony
Ransomware or logical corruption	SOC, SIEM	Crisis unit and security officer	To be defined depending on the extent	Last clean, immutable copy	Isolation; analysis; restoration from an immutable copy; posts in offline mode if needed
Power cut at a post	UPS	Automatic	No outage on UPS then generator	0	Switch to UPS and generator; clean shutdown if autonomy runs out

The RTOs and RPOs above are objectives; they are measured during drills and adjusted.

Who decides

Decision	Responsible
Automatic failover of a component	Monitoring (no human intervention)
Manual failover of a central site	WAW Telecom operations manager, after informing the DPAF
Activation of the crisis unit	Operations manager or security officer
Posts switched to prolonged degraded mode	DPAF (post chief, on instruction from management)
Restoration from site C	Crisis unit, with the DPAF's agreement
Communication to airlines and the Treasury	Crisis unit

Failover of a central site (standard procedure)

1. Confirm the outage (monitoring, health check, confirmation by the arbiter) and open a major incident.
2. Check that the replica at the surviving site is up to date (replication lag zero or known).
3. Promote the replica to primary (automatically or manually) and fence the former primary to avoid two simultaneous primaries.

4. Redirect traffic (DNS or virtual address) to the surviving site; check access to the four platforms and to the airline gateway.
5. Check the **Audit log** (**Chain intact**) and access to the HSMs of the surviving site.
6. Check the resynchronization of the post servers.
7. Inform posts, the DPAF, the DGCPT and airlines according to the communication plan.
8. Record each step and its time.

Switchback

1. Rebuild or repair the failed site.
2. Resynchronize it as a replica of the site that became primary.
3. Wait for stable synchronous replication.
4. Return to nominal operation during a planned maintenance window (deliberate switchback, never automatic).
5. Write the lessons learned and update the procedures.

Drills

- **At least two recovery drills per year**: actual failover from A to B, then back.
- One annual full restoration drill from site C.
- One annual link-cut drill at at least one land post.
- Each drill results in a report shared with the DPAF.

Runbooks to be written

- Database failover and switchback.
- DNS or virtual address failover.
- HSM replacement and key ceremony.
- Full restoration and point-in-time restoration.
- Rebuild from site C.
- Offline mode of a post and resynchronization.
- Ransomware response.
- Rotation of device and airline keys.
- Crisis communication.

5. Security and compliance

Area	Measure (Target)
Segmentation	Separate zones: Internet exposure (WAF, public web), API, data, administration, links to posts, links to partners; filtering between zones
Administration access	Single bastion per site, strong authentication (MFA), nominative accounts, session recording, time-limited privileges
User identity	OIDC, MFA and PKI officer cards; MFA is **To be developed** (the demonstrator uses username and password)
Devices	Replacement of the shared `WAWID_GATE_KEY` key with per-device certificates (mTLS), **To be developed**

Area	Measure (Target)
Security monitoring	24/7 SOC and SIEM, correlation of application, system, network and HSM logs
Vulnerabilities	Continuous scanning of images and systems, patches according to criticality, penetration tests before go-live and at least once a year
Audit log	SHA-256 chain (Existing) periodically anchored by an HSM signature (To be developed)
Personal data	CDP formalities before any production use; record of processing activities; procedure for access and rectification rights
Physical security	Access control for rooms and racks, CCTV, intervention log, locked racks for WawID equipment
Personnel	Clearance of people with privileged access (procedure **To be checked** with the authorities); confidentiality undertaking
Key ceremonies	Generation, backup and rotation of keys in the presence of witnesses, signed minutes, multiple custodians

6. Deployment phases

Durations are **indicative** and to be confirmed at the end of phase 0. They are longer than those of the proposal note (`docs/PROPOSITION-REMPLACEMENT-SECURIPORT.md`), which did not include building the high-availability infrastructure. Phases 1 and 2 run in parallel.

Phase	Content	Entry criteria	Exit criteria	Indicative duration
0. Scoping and legal	Contract; CDP formalities; agreement with Sénégal Numérique (site A) and with the commercial data center (site B); choice of site C; collection mandate for the security fee if the `OPERATEUR` flow is selected; agreements with the file holders	Launch decision	Agreements signed; CDP formalities initiated; architecture validated	1 to 3 months
1. Build of sites A, B and C	Racks, power, network, A ↔ B links, carriers, firewalls, WAF, HSMs, Kubernetes, PostgreSQL, object storage, backups, SOC	Phase 0 completed; hardware delivered	Infrastructure tests passed, including an A → B failover and a restoration from C	3 to 6 months
2. Application migration	Move from the Netlify demonstrator to the sovereign stack (see the table below)	Priority developments started	Functional acceptance in pre-production; load test; penetration test	3 to 6 months, in parallel with phase 1
3. AIBD pilot	A few counters and eGates at arrivals and departures, in parallel with the system in place; AIBD post server; gateway open to volunteer airlines; security fee reconciliation	Phases 1 and 2 validated; CDP formalities completed; officers trained	Pilot acceptance; no blocking incident during the observation period; failover drill passed in real conditions	2 to 3 months

Phase	Content	Entry criteria	Exit criteria	Indicative duration
4. Rollout in waves	Wave 1: full AIBD, ZIG, PDK. Wave 2: ROS, KID, KAR (offline mode essential)	Pilot review; post servers installed; links doubled	Acceptance per post, including a link-cut test; decommissioning of the previous system under the contractual clauses	1 to 2 months per wave
5. Operations (MCO)	24/7 monitoring, maintenance, drills, enhancements	Go-live report	Annual review	Ongoing

Migration from the demonstrator to the sovereign stack (phase 2)

Item	Demonstrator (Existing)	Target	State
API	Single function `netlify/functions/api.mts` routed on `/api/*`	Same code and same routes in a Node.js 22 container, behind a gateway; the local server `dev-server.mjs` already shows that the function runs outside Netlify	To be developed (adapter and container image)
Storage	Netlify Blobs through `netlify/lib/store.mts`: one collection = one JSON document rewritten on each write	PostgreSQL (transactions, tables partitioned by month) and object storage; `store.mts` replaced by an SQL repository; end of the global `/api/data` load	To be developed
Secrets	Netlify environment variables (`SESSION_SECRET`, `WAWID_GATE_KEY`, `WAWID_CARRIER_KEYS`, connector tokens)	Secrets vault and HSM; separate keys for sessions and for QR codes	To be developed
Static files	`public/` published by Netlify	Web tier in each site, behind the WAF, one domain per platform	To be developed (configuration)
Security headers	`netlify.toml`	Reproduced on the reverse proxy and the WAF	To be developed (configuration)
Demo dataset	Created automatically if the database is empty (`netlify/lib/seed.mts`)	Disabled in production	To be developed
Audit log	SHA-256 chain in storage	Append-only table, HSM anchoring, export to C	To be developed (anchoring)
Identity	scrypt password, signed 12-hour session	OIDC, MFA, PKI officer cards	To be developed
Offline mode	None	Post server with store and forward	To be developed

Item	Demonstrator (Existing)	Target	State
eGate kiosks	`/gate/` with the `X-Gate-Key` header	Served by the post server; per-device certificates	To be developed
NFC bridge	`tools/nfc-bridge.mjs`, origin allowed by `NFC_BRIDGE_ORIGIN`	`NFC_BRIDGE_ORIGIN` limited to production domains (`dpaf.<domain>`, kiosk domain)	Configuration
Mobile app	`mobile/`, default API `https://wawid-waw-telecom.netlify.app`	Built with `--dart-define=WAWID_API=https://api.<domain>`	Configuration
Airline gateway	`POST /api/carrier/messages`	Same contract, exposed on `compagnies.<domain>` behind the WAF	Configuration
Data	Fictitious	No takeover from the demonstrator; possible takeover from the system in place under the contractual clauses	To be checked

Acceptance tests

Test	Phase	Success criterion
Load test	2, then before each wave	10 checking transactions per second sustained and 50 in bursts; AIBD peak of 15,000 travelers per hour carried by **a single** site
Failover test A → B and back	1, 3, then twice a year	RTO ≤ 1 h; RPO met; audit log intact
Restoration test (pre-production and from C)	1, then every quarter	Full restoration successful; duration and loss measured
Penetration test	2, before the pilot, then every year	No open critical or high vulnerability
Link cut at a post	3 and 4	Checks continue; crossings resynchronized with no loss or duplicate
Loss of an HSM	1	No interruption of signing
Functional acceptance	2 and 3	Acceptance table of the *Installation guide — Senegal*

7. Environments

Environment	Location	Data	Use
Production	Sites A and B (active), C (vault)	Real	Operations
Pre-production	Site B, isolated from production	Synthetic or anonymized	Validation of releases, load and restoration tests, drills
Test and training	Site B or a dedicated environment	Fictitious	Acceptance, officer training, airline trials
Development	WAW Telecom	Fictitious	Development, `npm test`

Environment	Location	Data	Use
Netlify demonstrator	Netlify, outside the national territory	Fictitious only	Demonstrations; no real personal data

Each environment has its own secrets, keys and accounts. A release is deployed to production only after validation in pre-production.

8. Operations

Monitoring

- **24/7 network operations center (NOC)**, coupled with the SOC.
- Health checks of both sites, replication, each post's links, HSMs, backups and certificates.
- The status of post devices is also shown in **System monitoring** in the DPAF space (Existing for display; real reporting To be developed).
- Capacity and performance dashboards, alert thresholds and on-call duty.

Service levels to negotiate

These items are to be negotiated with Sénégal Numérique and the commercial data center; no value is set in this plan:

- availability of power and cooling;
- on-site intervention and emergency access times;
- availability and restoration time of carrier links;
- anti-DDoS protection and filtering capacity;
- physical security, access control and visitor log;
- notification of incidents and planned maintenance;
- reversibility and exit conditions;
- penalties for failures.

Capacity, patches and changes

- **Capacity**: quarterly review of actual load (crossings, e-Visas, API/PNR messages, biometrics) against the sizing; each site must be able to carry the peak alone.
- **Patches**: update site by site (B then A), without interruption thanks to active-active; critical patches applied urgently under the change procedure.
- **Changes**: change advisory board with the DPAF; planned and announced maintenance windows; every change is first applied in pre-production.
- **Planned maintenance**: preferably during off-peak traffic hours, announced in advance to posts, airlines and the Treasury.

Responsibility matrix (RACI)

R: responsible; A: accountable; C: consulted; I: informed. Proposed allocation, **To be checked** in the agreements.

Activity	WAW Telecom	DPAF	Sénégal Numérique	Commercial DC	DGCPT	Airlines
Hosting of site A (power, room, access)	C	I	R/A	—	—	—
Hosting of site B	C	I	—	R/A	—	—
WawID infrastructure and platform	R/A	C	C	C	—	—
Backups and restorations	R	A	C	C	—	—
Decision to fail over a site	R	A	I	I	I	I
Post operations (procedures, degraded mode)	C	R/A	—	—	—	—
Security (SOC, vulnerabilities)	R	A	C	C	—	—
CDP formalities	C	R/A	—	—	—	—
Security fee (settings, collections)	R	I	—	—	A	C
API/PNR transmission	C	A	—	—	—	R
Recovery drills	R	A	C	C	I	I

9. Bill of materials per site

Equipment types and quantities **to be sized** during the detailed study; no prices. The minimum quantities given follow from the redundancy principles.

Equipment	Site A	Site B	Site C	Each post
Racks	To be sized	To be sized	1 or 2	1 rack or cabinet
Routers (two carriers)	2	2	1 or 2	1 (+ 4G/VSAT router)
Firewalls	High-availability pair	Pair	1 or pair	1
Switches	At least a pair	At least a pair	1 or 2	2
Load balancers and WAF	Pair	Pair	—	—
Kubernetes servers	3 control-plane nodes + worker nodes to be sized	Same	—	—
PostgreSQL servers	At least 2	At least 2 (with separate pre-production)	—	—
Object storage	At least 4 nodes (erasure coding)	At least 4 nodes	Immutable storage to be sized	—

Equipment	Site A	Site B	Site C	Each post
Event bus	3 nodes	3 nodes	Possible arbiter	—
ABIS servers (biometrics)	According to supplier	According to supplier	—	—
HSM	2	2	Key backup (safe)	—
Backup	Backup server	Backup server	Immutable appliance + offline tapes or disks + safe	—
Bastion and monitoring	2 bastions, SIEM collectors	2 bastions, SIEM collectors	—	—
Post servers	—	—	—	2 nodes
Power	Provided by the host (To be checked)	Provided by the host	Depending on the site	UPS + generator

10. Risks and mitigations

Risk	Consequence	Mitigation
Certification or capacity of a data center not confirmed	Actual availability below the objective	Technical audit before signature; capacity clause; certified site B
Sites A and B too close (same area, same substation)	Simultaneous loss of both sites	Prefer a separate site B (Rufisque); check power feeds; distant site C
Cut of the A ↔ B link	Synchronous replication interrupted; risk of two primaries	Two fiber paths; arbiter at site C; manual failover if the arbiter is unreachable
Dependence on a single carrier	General network outage	Two carriers per site and per post
Delay of the posts' offline mode	Land posts stopped during an outage	Development priority; written manual procedure; land wave conditional on its validation
Ransomware	Encryption of production data and online backups	Immutable and offline copies; segmentation; administrator MFA; drills
Loss or compromise of keys	Unable to verify QR codes, restore or prove integrity	HSM pairs; split key backup; ceremonies
CDP formalities or security fee texts not completed	Production blocked	Started in phase 0; `TRESOR` flow usable pending the text for the `OPERATEUR` flow
Procurement lead times (HSMs, servers, fiber)	Delay of phases 1 and 4	Orders as soon as phase 0 ends; spare hardware
Insufficient operations skills	Poorly handled incidents	Training, written procedures, drills, skills transfer
Power cuts at posts	Equipment stops	UPS, generators, regular tests

Risk	Consequence	Mitigation
Load above assumptions	Degraded response times	×5 margin, quarterly review, horizontal scaling

11. Existing, target and to be developed

Topic	Existing (demonstrator)	Target	To be developed
Hosting	Netlify, outside the national territory	Sites A and B in Senegal + vault C	Building the sites
Storage	Netlify Blobs, one collection per JSON document	Replicated PostgreSQL and object storage	PostgreSQL storage layer
Availability	That of Netlify, without sovereign commitment	99.95% or more, two active sites	Infrastructure, procedures, drills
Backups	No dedicated backup	3-2-1-1-0 rule	Full implementation
Keys and secrets	Environment variables	HSM pairs, secrets vault	HSM integration
Identity	Password, 12-hour session	OIDC, MFA, PKI cards	MFA, OIDC
Posts	Browser connected directly to the central system	Post server, offline mode	Post server and resynchronization
Audit log	Verifiable SHA-256 chain	Chain anchored by the HSM, exported to C	Anchoring and export
Airline gateway	`POST /api/carrier/messages`; a resend creates a new message but merges the passengers of the same flight	Same contract, behind the WAF	Explicit idempotency key and resend guidelines
Security fee	`OPERATEUR` flow by default (`settings.redevance.circuit`); `TRESOR` flow available	Flow set by the texts	Real invoices; text and mandate if `OPERATEUR`
Retention	5 years displayed, no purge	Compliant purge and archiving	Purge job